

WHITE PAPER

Beyond the Single Ordinal Grade

The LENS × ARC framework for reporting intellectual quality and agency

A student's paper is usually compressed into a single number. That grade can look more exact than the marker's judgement permits, while saying little about the kind of thinking the student has demonstrated.

LENS × ARC describes the paper instead. It identifies the strongest intellectual move visible in the student's work and shows whether the student adopts, develops or challenges the available line of thought. The result is an appraisal at a resolution the evidence can reasonably support.

Lee McCulloch-James

August 2026

Framework: leelemacjames.github.io/qualitative-assessment-framework.html

Executive summary

DeepMark’s article, *GCSE grades are often a coin flip*, gives public force to a long-established problem in qualitative assessment: different competent examiners can award different marks to the same extended response. The article draws on Ofqual research reporting that agreement with a designated “definitive grade” is very high in mathematics and much lower in English and other essay-based subjects. Its headline is arresting, although “coin flip” compresses several distinct ideas: marker agreement, legitimate professional judgement, grade-boundary effects, and the validity of the scale itself.

Ofqual’s own formulation is more careful. In extended responses, adjacent marks may both be defensible. A difference from the principal examiner’s judgement therefore need not be an error. The assessment system still requires a single mark and then frequently converts it into a grade with hard boundaries. The resulting integer has greater apparent resolution than the underlying judgement. The proposed LENS × ARC framework responds by changing what is reported. It records two features of the performance:

- **LENS: intellectual quality**, expressed through the strongest visible move: *Lay out, Examine, Notice, Sift*;
- **ARC: intellectual agency**, expressed through the response’s relation to an available line: *Adopt, Rework, Challenge*.

The resulting two-letter description, supported by evidence and a realistic next move, offers a coarser and more interpretable account than a single grade. It also separates faithful assimilation from discernment and principled independence. Originality receives no automatic premium: unsupported departure appears as *LC, Contrarian assertion*, while intelligent judgement within a supplied frame can appear as *SA, Judgement within frame*. LENS × ARC is presented here as a design proposition requiring piloting, moderation and validation. It is a performance map, not a personality inventory, and should never become another disguised numerical ladder.

The provocation: when a grade behaves like a coin flip

DeepMark’s article opens with a sharp comparison. Drawing on Ofqual’s 2018 work, it reports that the probability of receiving the “definitive grade” was approximately 0.96 in mathematics and around 0.55–0.60 in English language, English literature and history [1, 2, 3]. At classroom scale, the contrast is vivid: far more English candidates than mathematics candidates receive a grade different from the one assigned by a principal examiner.

The phrase “coin flip” succeeds rhetorically because it makes uncertainty visible. It can also be misunderstood. The reported probability is not the probability that an examiner has guessed the student’s attainment correctly. Rather, it is the probability of agreement with a particular reference judgement. Ofqual explicitly observes that more than one mark may be plausible and equally legitimate for an extended response [3]. A mark of 5 and a mark of 6 can both represent defensible applications of a qualitative descriptor.

Three claims should therefore be separated:

1. **Exact agreement is limited.** Competent judges do not always assign the same mark to the same work.
2. **Some disagreement is legitimate.** Qualitative descriptors permit a bounded region of reasonable professional judgement.
3. **Hard reporting boundaries amplify small differences.** Adjacent defensible marks can become different grades with unequal consequences.

The first is a reliability problem. The second concerns the nature and warranted resolution of qualitative judgement. The third is a reporting and decision problem. Treating all three as examiner error obscures the more fundamental issue: the system reports a point even where the evidence supports a region.

Reliability, accuracy and resolution

Assessment discussions frequently use *accuracy* and *consistency* as though they were interchangeable. They are not. Let T represent the defensible region of judgement for a response and let m_j be the mark assigned by judge j . A conventional system often behaves as though there were a unique target m^* and measures disagreement by

$$e_j = m_j - m^*.$$

For qualitative work, the stronger model may be a set-valued target,

$$T = [m_{\min}, m_{\max}],$$

within which more than one judgement is warranted and/or the max and min are removed from the panel as per figure skating et al qualitative scoring. Exact repeatability can then understate validity: two markers may disagree while both remain inside T . Conversely, repeated agreement can overstate validity if all judgements share the same systematic bias.

This distinction matters when AI is introduced. DeepMark argues that human marking tends to scatter, whereas machine marking can be highly repeatable and systematically displaced. The image is useful. A tight cluster may be precise and still miss the bullseye. A wide cluster may average around the target while giving uncertain outcomes to individuals. Neither pattern is sufficient evidence of trustworthy assessment.

The problem also concerns scale resolution. A nine-point grade scale appears to support unit-sized comparisons. Yet if plausible professional judgements span adjacent marks or grades, the scale's engraved divisions are finer than its evidential resolving power. The result is a micrometer engraved on a rubber ruler: a sharp-looking reading produced by an elastic act of appraisal.

From better marking to better reporting

More training, clearer mark schemes, double marking and calibrated machine assistance can improve agreement. These are worthwhile responses. They leave a prior question unanswered: *what information should the assessment report?*

A single grade collapses several educationally different performances. A response may reproduce a taught account fluently, apply a supplied analytical method, notice a neglected implication, compare plausible explanations, or reject a frame with proportionate evidence. Two responses can receive the same grade while exhibiting different intellectual qualities. Two adjacent grades can arise from performances that are educationally indistinguishable at the available level of resolution.

LENS × ARC therefore changes the object of reporting. Its purpose is descriptive before it is classificatory. It rather asks:

What is the strongest intellectual move visible in this response, and how does the response handle the supplied, taught or shared line of thought?

The framework reports broad regions of performance, cites the evidence for the judgement, and identifies a plausible next move. It makes fewer claims than a finely divided numerical scale and aims for greater interpretability.

The LENS × ARC architecture

The matrix pits faithful reconstruction towards increasingly discriminating judgement against increasing ability to express supportable challenging views.

LENS of quality

LENS identifies the strongest move demonstrated in the piece of work:

L: Lays out

Reproduces available content or reasoning.

E: Examines

Makes parts, relations and warrants explicit.

N: Notices

Perceives a non-obvious feature or implication.

S: Sifts

Discerns what matters among alternatives.

The sequence moves from faithful reconstruction towards increasingly discriminating judgement. It should not be read as an equal-interval scale. A response can contain several kinds of move, and the code records the strongest one for which sufficient evidence is visible.

ARC of agency

ARC describes how the response handles an available line:

A: Adopts

Follows an available line.

R: Reworks

Extends a shared line.

C: Challenges

Tests or departs from the line.

ARC is relational. A response cannot be judged as adoptive, reworking or challenging in the abstract; the judgement depends upon what was supplied, taught, prompted or shared. The same response may therefore occupy a different ARC position under a different task design.

The matrix

LENS	A: Adopt	R: Rework	C: Challenge
L: Lay out	LA Accurate echo	LR Elaborated echo	LC Contrarian assertion
E: Examine	EA Applied analysis	ER Developed analysis	EC Independent analysis
N: Notice	NA Prompted perception	NR Original noticing	NC Reframing insight
S: Sift	SA Judgement within frame	SR Comparative judgement	SC Principled independence

The matrix prevents agency from masquerading as quality. Challenge is not intrinsically superior to adoption. *LC* records departure without adequate evidence or reasoning, while *SA* records intelligent weighing within supplied criteria. Equally, fluent assimilation is not dismissed as mindless repetition: accurate reconstruction and applied analysis are genuine achievements, described at the resolution demonstrated.

Assimilation, discernment and intellectual independence

The framework can be understood as a response to a familiar distortion in assessment. Schools often reward successful assimilation and describe it as independent analysis. A student adopts the vocabulary, structure, criteria and acceptable inferences of a discipline, then reconstructs them fluently. This is valuable apprenticeship. The misdescription begins when faithful reconstruction is treated as evidence of autonomous judgement.

LENS × ARC makes the distinction inspectable:

- **Assimilation** concerns the capacity to reconstruct and work within inherited thought.
- **Discernment** concerns the capacity to notice what is consequential, compare plausible alternatives and judge what carries weight.
- **Intellectual independence** concerns warranted revision or rejection of the available frame.

These capacities interact without collapsing into one another. Strong assimilation can coexist with limited discernment. A learner may also glimpse an important exception without possessing the analytical resources to warrant it. Principled independence requires both discrimination and accountability to evidence.

The two-letter codes make these distinctions memorable and may superficially resemble a personality typology. That resemblance should stop at the mnemonic. Myers–Briggs-style instruments purport to classify persons into relatively stable types. LENS × ARC classifies the evidence in a particular performance under particular conditions.

A performance map, not a personality inventory. “This response demonstrates ER” is a warranted claim when evidence is cited. “This student is an ER thinker” exceeds the evidence and converts a provisional description into an identity.

Reporting practice

The recommended reporting unit contains four elements:

LENS + ARC + evidence + next move

For example:

ER: Examine × Rework. Develops the shared account by making one significant causal relationship explicit.

Next move towards SR: sift that explanation against a plausible alternative and state why one deserves greater confidence.

This format makes the evidential warrant visible and retains the formative function of assessment. The code alone is insufficient. Without the cited move, it risks becoming another opaque grade. Without the next move, it describes performance while offering little direction for development.

Insufficient evidence sits outside the matrix. A blank, fragmentary or withheld response supports no claim about intellectual quality. The appropriate record is that the quality was not demonstrated in the available evidence. Inability, indifference and disengagement are separate hypotheses requiring separate evidence across time.

What LENS × ARC claims, and what it does not

The framework makes four modest claims:

1. broad qualitative regions may match the resolution of extended-response evidence better than narrowly divided marks;

2. separating intellectual quality from relation to an available line produces more interpretable feedback;
3. evidence-linked descriptors can support moderation around the kind of thinking demonstrated;
4. reporting a next move can connect summative judgement to learning.

It does not yet establish improved inter-rater reliability, predictive validity, freedom from demographic bias, or suitability for high-stakes certification. Nor does it abolish judgement. It relocates judgement into broader, more explicit categories and asks markers to expose their warrant.

The framework should also remain distinct from a hidden score. If institutions map LA through SC onto a twelve-point ladder, the original problem simply returns under new labels. The matrix is not unidimensional: movement rightward can produce LC, and adoption can coexist with high-quality sifting. There is no defensible total ordering of all twelve cells.

A programme for piloting and validation

A responsible pilot should examine both reliability and educational usefulness.

Phase 1: descriptor testing

Markers independently code a varied sample of responses, cite the textual move supporting each code, and record uncertainty between neighbouring regions. Cognitive interviews identify terms that invite inconsistent interpretation. Particular attention should be given to the boundaries between Examine and Notice, and between Rework and Challenge.

Phase 2: comparative moderation

The same responses receive both conventional grades and LENS × ARC reports. Researchers compare:

- exact and adjacent agreement on each axis;
- time required to reach and explain a judgement;
- the frequency with which disagreement is resolved by inspecting evidence;
- whether teachers and students interpret the report consistently;
- whether codes remain stable when task prompts change.

Agreement should be reported separately for LENS and ARC. A single combined agreement coefficient can conceal which axis is functioning poorly. Weighted statistics may be appropriate for the ordered LENS categories, while ARC requires caution because Adopt, Rework and Challenge are relations rather than equal steps on a universal scale.

Phase 3: consequential validity

Students receive conventional feedback, LENS × ARC feedback, or both. Subsequent work is examined for the quality and feasibility of the next move. Interviews should test whether the framework enlarges students' understanding of intellectual work or encourages strategic code-chasing. Differential patterns by subject, language background, disability, gender and other relevant groups should be investigated before consequential use.

Phase 4: calibrated human and machine assistance

AI may help locate candidate evidence, compare a response with descriptors, or provide a consistent first pass. Human responsibility remains essential, particularly where context determines whether a move genuinely challenges an available line. Evaluation must distinguish repeatability from agreement with expert judgement and from educational validity. Machine confidence should never substitute for an evidential citation.

Policy implications

DeepMark’s argument points towards more consistent marking support. LENS × ARC points towards more honest reporting. The approaches can complement one another if their purposes remain distinct.

For classroom and institutional use, five principles follow:

1. **Match reported precision to evidential resolution.** Do not imply unit-level certainty where professional judgement supports a region.
2. **Preserve the work sample.** A code acquires meaning through the move cited from the response.
3. **Separate consistency from validity.** A tightly clustered judgement can remain systematically misplaced.
4. **Keep intellectual agency relational.** Record what account, prompt or frame was available to the learner.
5. **Protect against reification.** Use codes to describe performances, never to type students or rank all forms of thinking on one concealed scale.

For high-stakes qualifications, LENS × ARC is best understood initially as a reporting supplement and research programme. Replacing grades would require evidence of comparability, fairness, public interpretability and manageable moderation at scale. Classroom adoption can begin more modestly through selected extended responses, departmental moderation and student reflection.

Conclusion

The strongest lesson from the DeepMark provocation is not that examiners are careless or that every English grade is arbitrary. Extended-response assessment asks human judges to convert complex performances into sharp points on a narrow scale. Ofqual’s evidence shows that plausible professional judgements can straddle those points and their consequential boundaries.

LENS × ARC offers a different bargain. It sacrifices spurious numerical sharpness in exchange for a coarser, richer and more inspectable description. It records what the thinking does, how it handles inherited thought, what evidence warrants that judgement, and what development might follow.

A single grade answers, with doubtful precision, *how much?* LENS × ARC asks the more educationally useful questions: *what kind of intellectual move is visible, how independently is it warranted, and what should become possible next?*

References

- [1] G. Waugh, “GCSE grades are often a coin flip,” *DeepMark*, 19 August 2026. [Article online](#).
- [2] Ofqual, *Marking consistency metrics: an update*, Ofqual/18/6449/2, November 2018. [Report online](#).
- [3] B. Black, “11 things we know about marking and 2 things we don’t yet,” *The Ofqual Blog*, 5 March 2019. [Article online](#).
- [4] L. McCulloch-James, *Beyond the Single Grade: The LENS × ARC Framework*, 2026. [Framework online](#).